MICROCHIP
intel

Programmer PCB

A Custom 32-bit, 64-Pin PIC32MX MCU Compatible PCB for USB Bootloading, Firmware Flashing & Hardware Validation

ASU Arizona State University

BACKGROUND & GOALS

MCU FIRMWARE

MCUs (microcontroller units) are COTS (commercial off the shelf) ICs (integrated circuits) functioning on a PCB as a single chip computer with a processor, memory, I/O peripherals for external PCB hardware - sensors, motors, displays, LEDs.. MCU's come with blank memory. To use the MCU we must first write **bootloader firmware** to the internal flash memory using a **programmer.**


64-Bit MCU: PIC32MX534F064H-I/PT

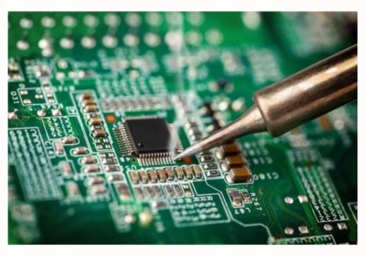

Bootload Firmware Example


MPLAB Snap In-Circuit Debugger PG164100


MPLAB PICKIT 4 PG164140

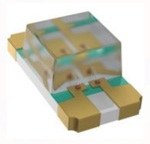
PROBLEM

Traditional MCU firmware testing depends on an assembled target board, where the MCU is soldered in place and programmed through an additional external header. This **limits flexibility** during bootloader development; hardware mistakes may require rework, replacement, or a new board revision - **ramping up cost** and **time** to manufacture.

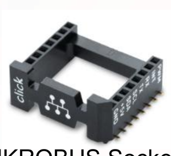



64-Pin Clamshell Socket


1x8 Programmer Socket


LEDs


USB Connector


MIKROBUS Socket

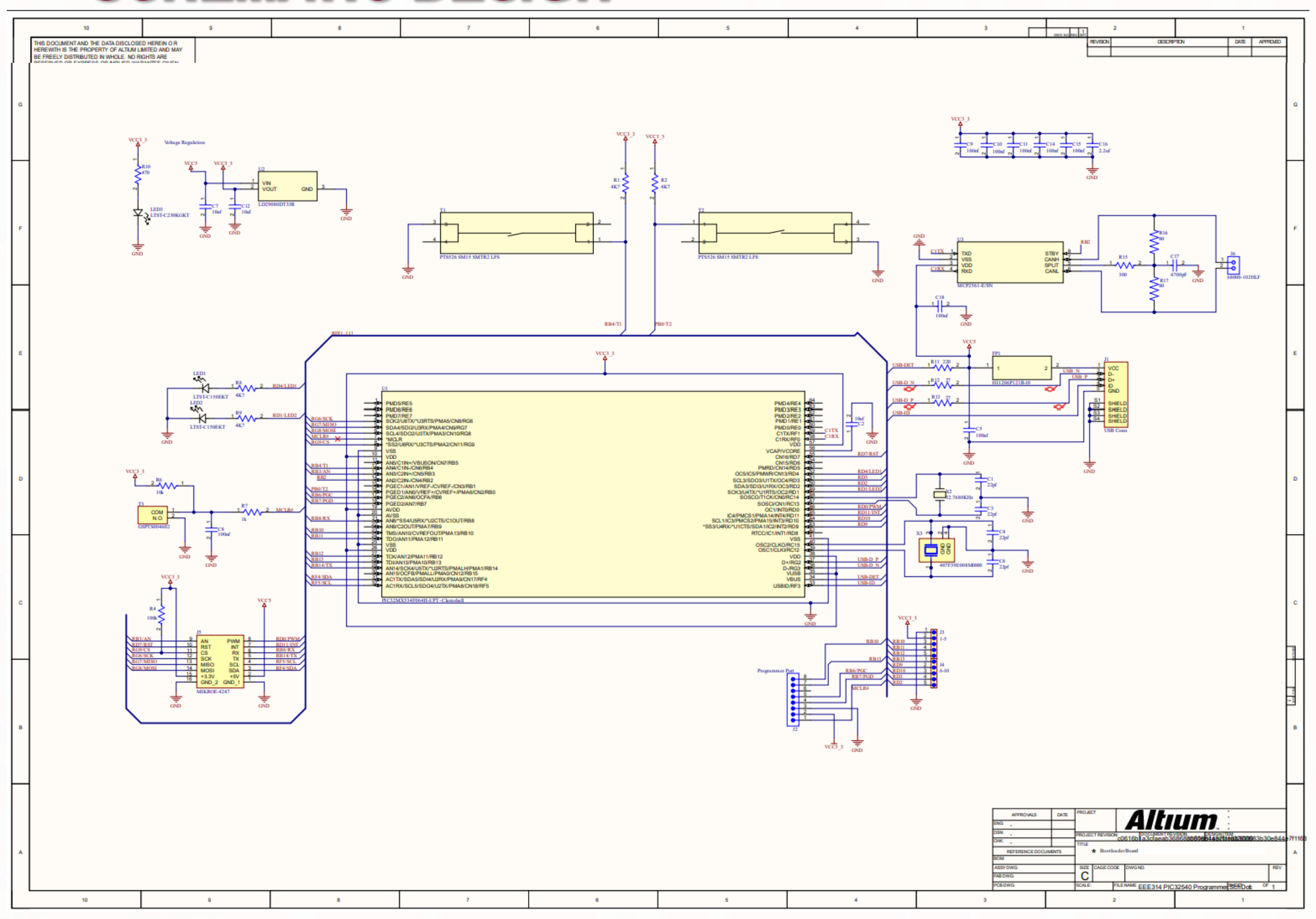
GOAL

Create a novel PCB bootloader, firmware flashing & hardware testing platform compatible with any PIC32MX 64-pin MCU, ready for use in research and industry.

A **reusable** socketed bootloader board with **modularity** for any PIC32MX 64-pin MCU. The clamshell socket allows the MCU to be powered, programmed, and tested without soldering directly to a PCB. Supporting ICSP firmware flashing, USB bootloading, reset/MCLR validation, peripheral breakout, CAN communication and hardware verification.

METHODOLOGY / PROCESS

SCHEMATIC DESIGN



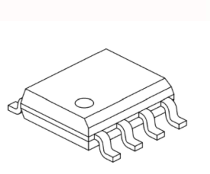
DESIGN INPUTS

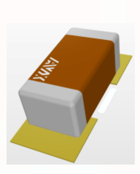
PIC32MX534F064H 64-pin MCU requirements
ICSP programming: MCLR, PGEC, PGED, VCC, GND

USB bootloader support: D+, D−, VBUS, USB-ID
Required support circuits: VCAP, oscillator, reset, decoupling
Expansion/testing: CAN, LEDs, buttons, headers, MikroBUS

PROCESS

1. Learn Altium Designer
2. Schematic Design
3. Component Selection
4. PCB Layout
5. PCB Ordering
6. PCB Assembly
7. PCB Test / Validation

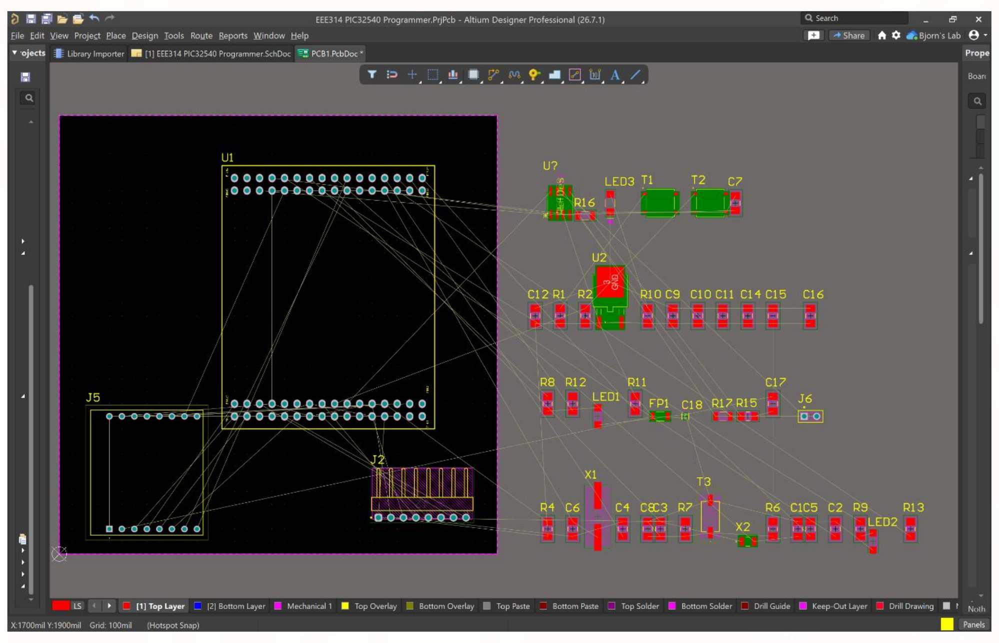




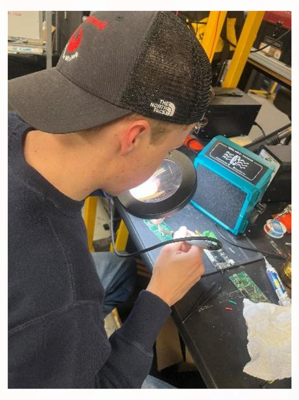
CHALLENGES

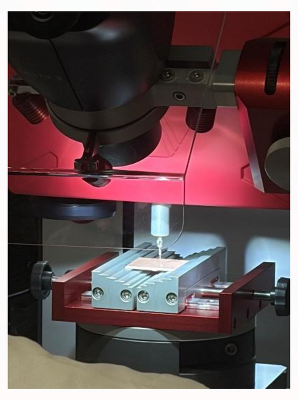
64-pin MCU -> socket pin mapping
USB differential pair D+/D− routing
3D Socket and header clearance
Signal integrity (SIG, GND, GND, SIG)
DRC errors 130 -> 18
JLCPCB manufacturing rules
Firmware test validation

PCB LAYOUT



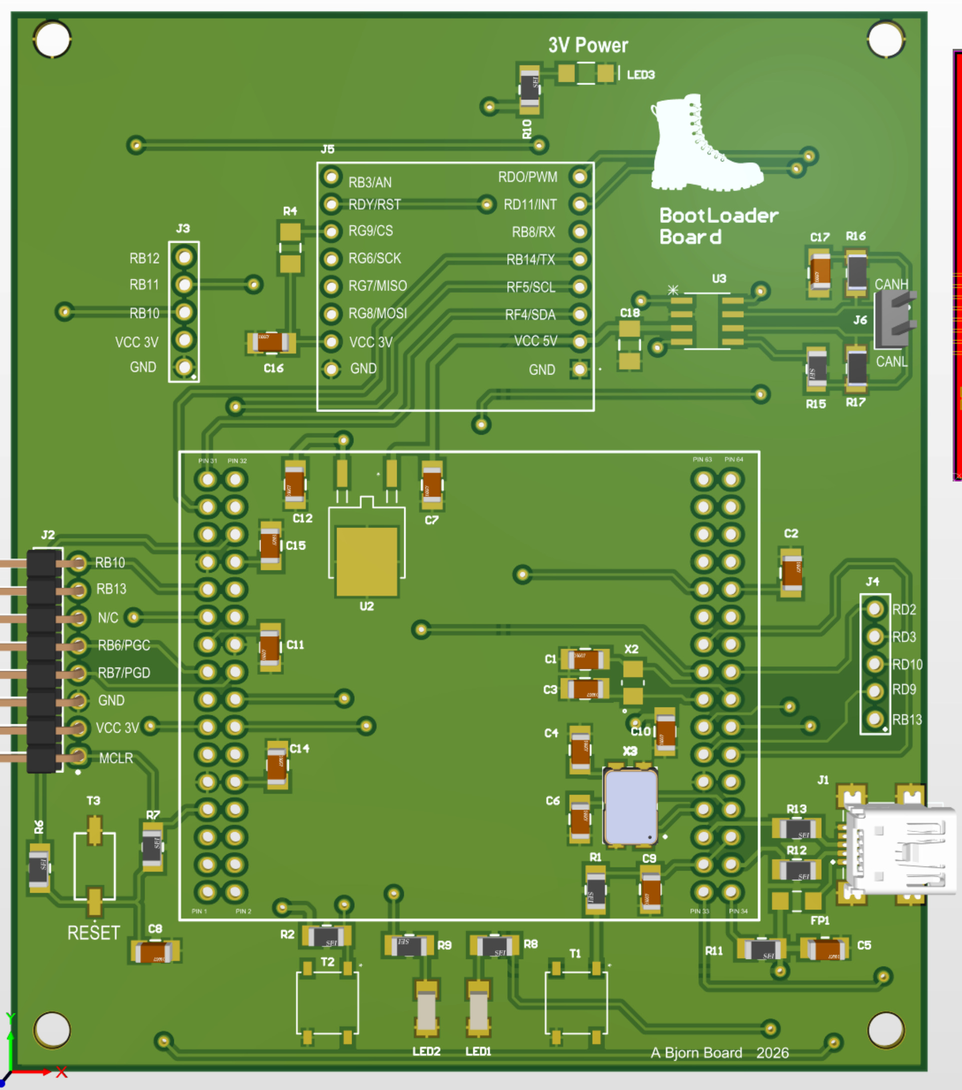
PCB ASSEMBLY, TEST, & VALIDATION

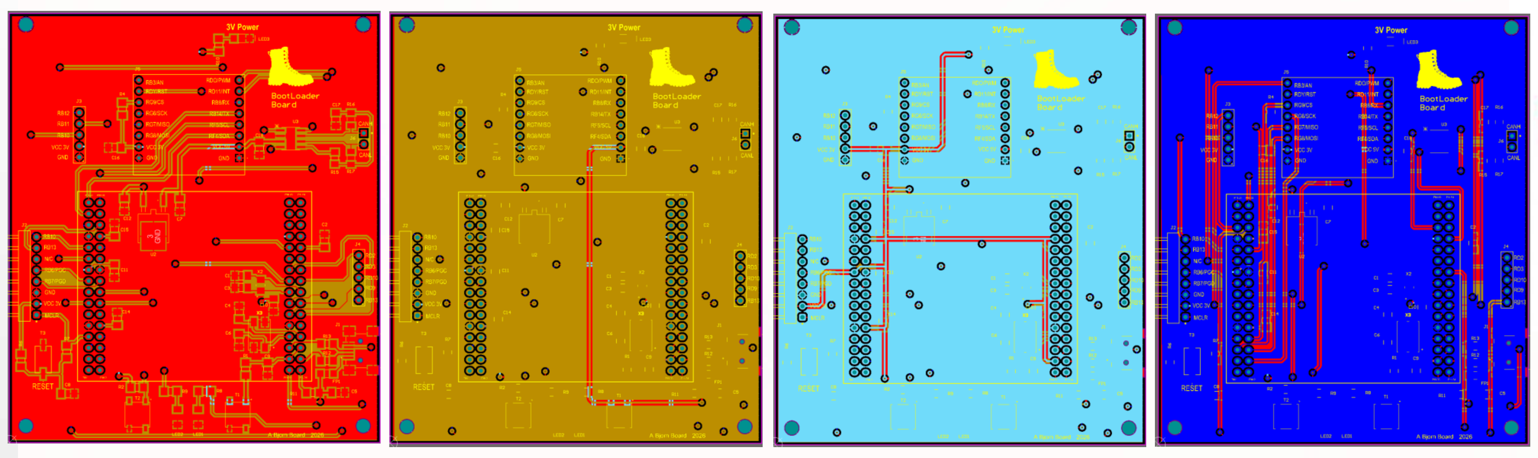


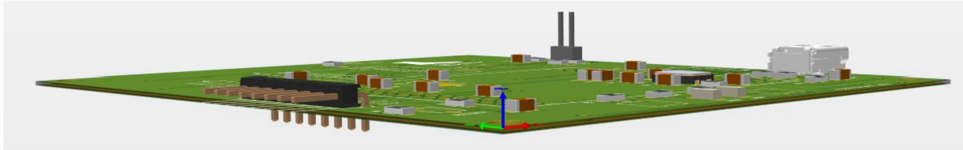


RESULTS / RESEARCH UTILITY

FINAL PRODUCT DESIGN



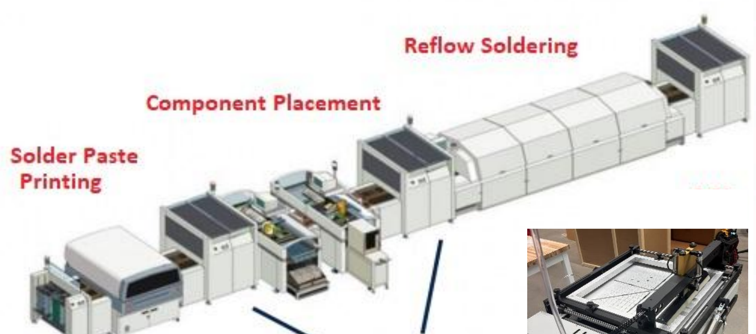




RESEARCH, INDUSTRY, & MARKET

Socketed programmer boards help industry teams reduce rework by testing firmware and hardware before committing MCUs to final assemblies.
The target users include embedded-system developers, university labs, PCB design courses, research teams, and low-volume electronics manufacturers.
The board provides a practical tool for firmware flashing, bootloader validation, and hardware testing during early product development.

ASU POLYTECHNIC SMT



To be manufactured in ASU POLY's SMT (surface mount technology) line in support of advanced semiconductor packaging research.

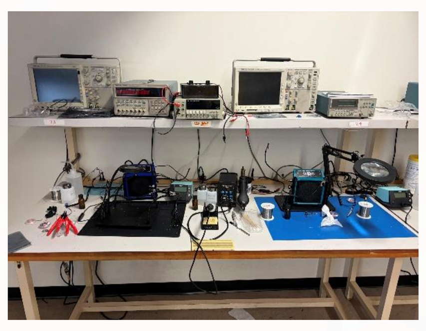

Solder paste printing


Component placement


Reflow soldering

ASU EEE 314

To be used Fall 26' in classroom for educational support of students and teachers in PCB design



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Bjorn Bengtsson